Electricity Revision Sheet 3



1. In your own words describe electrical:
    a. Current:

    b. Voltage:

    c. Resistance:

2. What are each of these measured in:
    a. Current:

    b. Voltage:

    c. Resistance:

3. If the Voltage did not change but the Resistance got bigger what would happen to the Current?

4. Calculate the missing numbers in these situations:
    a. Car horn:
        i. Voltage: 12 Volts
        ii. Resistance: 24 Ohms
        iii. Current: ____ Amps

    b. Car starter motor:
        i. Voltage: 12 Volts
        ii. Resistance: ____ Ohms
        iii. Current: 48 Amps

    c. Generator:
        i. Voltage: ___ Volts
        ii. Resistance: 10 Ohms
        iii. Current: 24 Amps

    d. Fridge:
        i. Voltage: 240 Volts
        ii. Resistance: ____ Ohms
        iii. Current: 2 Amps

5. How do objects get static charges? What happens when they discharge?

6. Describe the energy transfers for the following:

   a. Light bulb:

   b. Radio:

   c. Kettle:

   d. Hairdryer:

7. Why are parallel circuits better than series circuits? (2 reasons)

8. What is needed for a complete active circuit?

9. Draw an electrical diagram showing a single circuit with a light bulb, switch, battery and wires. Use the correct electrical symbols.

10. What things do you need to make:

    a. A very simple battery?

    b. An electromagnet?

11. List 4 sensible electrical safety rules for the home:

12. The following things make electricity safer to use in the home. Describe how each works.

    a. A fuse:

    b. An earth wire and pin:

    c. Insulated coatings around the wires: